

**11** Kate was asked to factorise  $x^2 + 5x + 6$  in the form  $(x + a)(x + b)$

Kate says,

“The sum of  $a$  and  $b$  must be 6 and the product of  $a$  and  $b$  must be 5”

(a) Explain what is wrong with Kate’s statement.

(1)

(b) Factorise fully  $2m^2 - 2$

(2)

(c) Factorise fully  $ax + bx - ay - by$

(2)

---

**(Total for Question 11 is 5 marks)**